DenoiseOpt: Residual Scoring and Bi-Level Meta-Learning for Wavetable Wrap Crackle

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Synth: github.com/reeldemo/reelsynth
Meta artifacts: github.com/reeldemo/denoise-opt-meta

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Synced with denoise-opt-meta `paper/v2`

Abstract

Open wrap seams inject audible crackle under cyclic wavetable playback. DenoiseOpt is a seam-local bake stack scored without paired clean labels. The outer meta objective is a prolonged residual score $\in [0, 1]$ $(1 = \text{best})$: tiled DenoiseOpt output vs an ideal multi-period reference (same seed, no open-wrap). Nested unsupervised loss $L = (1 - \mathcal{D}) + \lambda(1 - \mathcal{S})$ optimizes θ/λ inside each trial. Across 1500 trials, Meta Top 1 `evo_explore_515` reaches residual ≈ 0.824 vs DualCosine ≈ 0.698 (N=2000). Canonical PDF and versioning: denoise-opt-meta `paper/v2/`.

1 Pointer

Full paper, figures, and reproduction commands live in the public meta repo. This tree mirrors the latest residual figures for local builds.

```
cargo run -p reelsynth --release --bin bench_denoise_meta
python brand/artifacts/render_benchmark_matrix.py
```

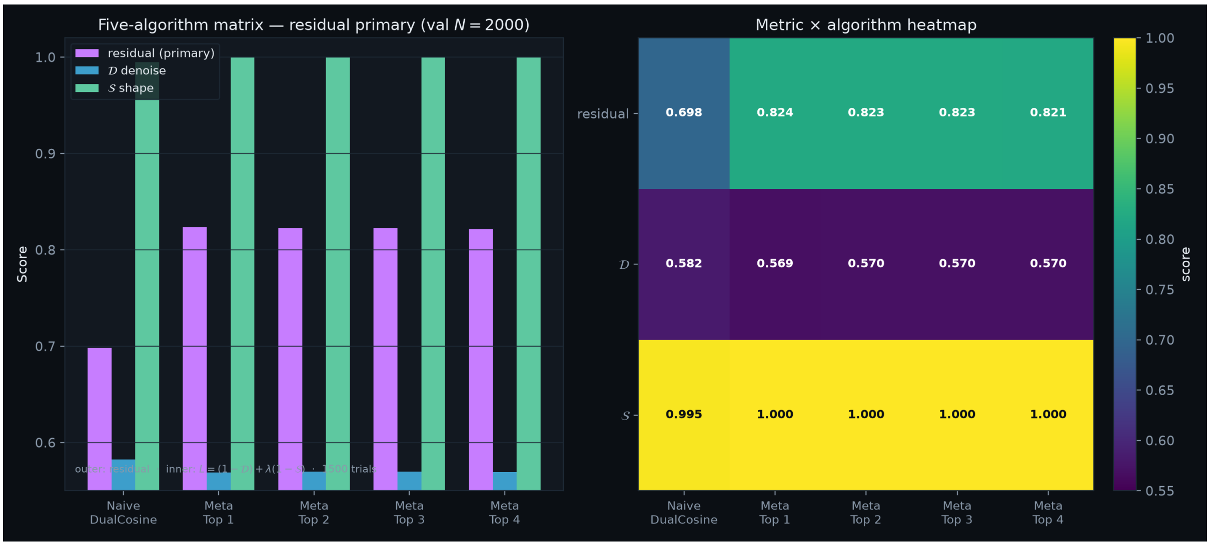

Figure 1: Residual-primary five-algorithm matrix (val N=2000).

Table 1: Headline residual scores.

Algorithm	Residual	Q
Naive DualCosine	0.698	0.789
Meta Top 1 (`evo_explore_515`)	0.824	0.784